

Design Multi-Protocol Implementation Plan

Protocol Integration

Throughout this course, you've explored three distinct communication protocols: MCP for standardized communication, A2A for task coordination, and ACP for collaborative execution. Now it's time to integrate this knowledge by designing a comprehensive multi-protocol implementation plan that leverages the strengths of each protocol.

Objective

The goal isn't just to describe three protocols—it's to craft a multi-protocol plan that uses MCP for standardized discovery/communication, A2A for fast, reliable task coordination, and ACP for collaborative execution in complex workflows. Are you mapping clear handoffs ($\text{MCP} \rightarrow \text{A2A} \rightarrow \text{ACP}$) across onboarding, rebalancing, and compliance? Are you designing for performance, security, and scale so diverse agents deliver real-time, regulated wealth management at enterprise depth?

Scenario

You're the chief AI architect for a large financial services company launching a new comprehensive wealth management platform. The platform needs to coordinate multiple types of AI agents

- Client Analysis Agents (analyze client financial profiles and risk tolerance)
- Market Research Agents (monitor market conditions and investment opportunities)

- Portfolio Optimization Agents (create and adjust investment portfolios)
- Risk Assessment Agents (evaluate and monitor investment risks)
- Compliance Agents (ensure regulatory compliance across all activities)
- Customer Service Agents (provide personalized client support and recommendations)

The platform needs to handle complex workflows like onboarding new clients, rebalancing portfolios, responding to market changes, and maintaining regulatory compliance—all while providing personalized, real-time service to thousands of clients.

Instructions for learners

Your Task



Design a comprehensive multi-protocol implementation plan that strategically uses MCP, A2A, and ACP protocols to create an effective, scalable, and secure wealth management system.

Complete these steps

1. Protocol Selection and Justification

For each major system function, identify which protocol(s) should be used and why:

- How would you use MCP for standardized communication across diverse agent types?
- Where would A2A protocols be most valuable for task coordination?
- Which scenarios require ACP's collaborative execution capabilities?
- Justify your choices based on the specific requirements of each function

2. Integration Architecture Design

Design how the three protocols will work together:

- Map out the communication flow between agents for a typical client onboarding process
- Describe how protocols hand off to each other (e.g., MCP discovery leading to A2A coordination leading to ACP orchestration)
- Design fallback mechanisms for when one protocol fails

3. Performance and Security Considerations

Address enterprise-scale requirements:

- How will you optimize performance across multiple protocols?
- What security measures are needed for each protocol in this financial services context?
- How will you monitor and measure the effectiveness of your multi-protocol system?
- Describe your approach to scaling the system as the client base grows

Completion Criteria:

- Clear protocol selection with justified reasoning for each use case
- Comprehensive integration architecture showing protocol interactions
- Practical consideration of performance, security, and scalability requirements

Conclusion

Submit your implementation plan as a structured document addressing all three sections. Your plan should demonstrate understanding of each protocol's strengths and appropriate application scenarios, while showing how they can work together effectively in a complex enterprise environment.